Sami Koneru-Ansari

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Education

University of California, Berkeley — **May 2028 (Expected)**
Bachelor of Science, Electrical Engineering and Computer Science — **4.0 GPA**

- **Relevant Coursework**: Data Structures and Algorithms, Machine Learning, Machine Structures, Signal Processing, Computer Networks, Discrete Mathematics, Probability Theory, Linear Algebra

Experience

Intel — **Sept. 2026 – Present**
AI Software Engineer Intern — *Folsom, CA*

- Developing deep learning primitives in **oneDNN**, Intel's open-source library behind PyTorch and TensorFlow on Intel hardware, working in **C++ on Linux** across **floating-point arithmetic**, numerical accuracy, and library architecture.

Piris Labs — **June 2026 – Aug. 2026**
Machine Learning Engineer Intern — *Remote*

- Built ML infrastructure to optimize **LLM inference** on **B200 GPUs**, spanning automated GCP deployment, **SGLang** serving, prefix caching, multi-tenancy, and per-instance routing, as groundwork for custom photonics hardware.
- Researched and implemented **scheduling and batching policies** to decrease median E2E latency by **over 50%**.
- Benchmarked speculative decoding, batching, scheduling, and routing policies on unified and disaggregated systems to optimize unit economy and throughput, achieving **483 tok/s** on GLM 5.2, **23% more** than Databricks.

TokenWorks — **Feb. 2026 – May 2026**
Data Science Intern — *Remote*

- Fine-tuned **BART-base** as a sequence-to-sequence tagger for low-resource Sumerian, jointly predicting part of speech, lemma, and English gloss from **217K CoNLL-U tokens** of transliterated cuneiform literature, in a single decoding pass.
- Built a self-hosted **Wikibase lexical knowledge graph** with idempotent Python importers mirroring Wikidata's Sumerian lexeme schema through a resumable, checkpointed pipeline, pulling **∼14,000 tokens** from Wikidata.

GetGreen — **Sept. 2025 – Dec. 2025**
Machine Learning Engineer (Contract) — *Remote*

- Engineered a **RAG chatbot** using LangChain and ChromaDB for semantic search and retrieval over **1,000+ scraped articles** processed with BeautifulSoup, pandas, and SQL, answering user sustainability questions in **under 3 seconds**.

California Department of Technology — **May 2024 – July 2024**
Software Development Intern — *Rancho Cordova, CA*

- Developed **full-stack web apps** with HTML, CSS, and JavaScript on the frontend, Node.js on the backend, PostgreSQL for relational data, and **JWT** session authentication to manage scheduling and file sharing for **100+ employees**.
- Managed datasets of **500+ employees** and **70+ clients** using SQL and Excel to drive operational reporting.

Projects

Gated Delta SSM | *Python, PyTorch*

- Implemented a **DeltaNet** write gate conditioned on the memory already stored at the target address, read from two scalars the delta rule computes anyway, cutting loss **5.5%** across 3 seeds for zero extra state and **+0.9% FLOPs**.
- Outperformed an oracle baseline handed ground-truth class labels (**5.67% vs 5.42%** loss reduction) on a task where token identity carries no signal, since that oracle learns one write strength per class while the gate adapts per token.

Catan Bot | *Python, PyTorch, Gymnasium*

- Trained an **actor-critic transformer** on a 512-float observation and 512-action masked space in an **AlphaZero-style** loop, distilling **PUCT** tree-search visit counts into both the policy prior and the value net after every self-play round.
- Built a **Playwright** client that intercepts colonist.io WebSocket traffic and maps live game frames into the agent's observation space through an event-driven state parser, built against a **3,700-frame** capture of production traffic.

Custom Digit Classifier | *C++, CUDA*

- Engineered and trained a **Convolutional Neural Network** from scratch in **C++/CUDA**, hand-writing a training and backpropagation pipeline for convolution, pooling, and dense layers with Adam optimization and no cuDNN.
- Wrote GPU-optimized **CUDA kernels** implementing **im2col and GEMM**, reaching over **99% accuracy** on the MNIST dataset and running at **∼50% the throughput** of an equivalent PyTorch model for MNIST on T4 GPUs.

Technical Skills

Languages: Python, C, C++, Java, SQL, Rust, JavaScript, TypeScript, Scheme, Bash, CUDA, RISC-V Assembly
Libraries: PyTorch, HF Transformers, Gymnasium, NumPy, pandas, Matplotlib, BeautifulSoup, Flask, LangChain
Tools: Git, Docker, Kubernetes, AWS, SGLang, Ollama, GCP, Linux, Playwright, PostgreSQL, ChromaDB, Supabase